

The final panel answers the operational question: *what total uncertainty attaches to a reported subgroup mean SGPC, and how stable are resulting classifications?*

For each of five representative subgroups (at $N = 500$, independent design), Panel E displays nested uncertainty layers:

- • **Truth** (amber dot): θ_s^{true} from linked data.
- • **Bridge estimate** (black dot): $\hat{\theta}_s^{\text{bridge}}$. The **arrow** between truth and estimate is the bridge bias (accuracy cost).
- **Sampling band**: 95% CI from repeated sampling (± 3.5 SGPC).
- **Stress-test band**: additional sensitivity to copula family choice, $P \perp U$ relaxation, and classification perturbation.

The total uncertainty envelope combines:

$$\underbrace{\text{bridge bias}}_{\text{Panel B}} + \underbrace{\text{sampling precision}}_{\text{Panel D}} + \underbrace{\text{model sensitivity}}_{\text{stress tests}}.$$

Action thresholds and category labels are meaningful only within this combined envelope. A subgroup classified as “low growth” at $\hat{\theta}_s = 35$ is robust if the full band stays below 40; it is fragile if the band spans 30–45.⁵